A Deep Transfer Learning Method for Estimating Fractional Vegetation Cover of Sentinel-2 Multispectral Images

Ruyi Yu, Shanshan Li, Bing Zhang, and Hongqun Zhang

***Abstract*— Fractional vegetation cover (FVC) is an important indicator for exploring hydrosphere, pedosphere, atmosphere, biosphere, and their interactions. Deep learning (DL) is a potential tool to handle large-scale data and approximate the complex nonlinear relationship between variables. It is, therefore, suitable for FVC estimation. However, few DL-based algorithms have been developed to estimate FVC as it is difficult to obtain a large amount of training data. This letter presents a novel method by means of deep transfer learning to address this issue. The proposed technique consists of two steps. In the first step, a large amount of simulated training samples were generated by a physical model (PROSPECT + SAIL radiative transfer model, PROSAIL). In the second step, a long short-term memory (LSTM) network was pretrained with the simulated training dataset obtained in the first step. Then limited real samples from satellite images were used to fine-tune the pretrained network. Experiments were conducted for the Sentinel-2 multispectral satellite images of two areas and the results were compared with those obtained by the traditional the Normalized Difference Vegetation Index (NDVI)-based method and two machine learning approaches. The results demonstrate that the performance of our method outperforms other advanced FVC estimation methods.**

***Index Terms*— Fractional vegetation cover (FVC), long short-term memory (LSTM), PROSAIL, transfer learning (TL).**

I. Introduction

FRACTIONAL vegetation cover (FVC) is defined as the percentage of the vertical projected area of green vegetation (including leaves, stems, and branches) to the total statistical area [1], [2]. It is a crucial parameter for delineating vegetation status, which is frequently used to characterize vegetation changes caused by climate and environment [3]. Generally, there are three main FVC estimation methods using remotely sensed data, including empirical models, pixel unmixing models, and physical model-based methods [4]. The empirical models are based on the statistical relationship between FVC and spectral band reflectance, or vegetation index (VI) from airborne or satellite spectra [5]. Empirical methods rely on *in situ* measurement data in specific regions. Therefore, the models were only suitable for specific vegetation types and regions, and if expanded to a large scale, these models may be invalid. The pixel unmixing models assume that each pixel is composed of several components and consider the proportion of vegetation components as the FVC of the pixel [6]. The challenge of this method is that it is difficult to accurately determine the endmembers. The physical model-based methods are based on inversion of canopy radiative transfer models that simulate the physical relationships between vegetation canopy spectral reflectance and FVC. Because of the complexity of the physical models, direct inversion is generally complex. Lookup table (LUT) methods and machine learning methods, such as back propagation neural networks (BPNNs), random forests (RFs), and Gaussian process regression (GPR), are widely used for indirect inversion of physical models [4], [7], [8]. Deep learning (DL), as an advanced machine learning technique, can approximate complex nonlinear relationships between environmental variables owing to multilayer learning [9]. However, few studies have applied DL to the inversion of FVC. This is because the performance of the DL networks relies on a large number of training data, but it is difficult to obtain sufficient ground truth data of FVC. Transfer learning, which tries to transfer the knowledge from the source domain to the target domain, is an important tool to reduce the requirement for a large training dataset in the target domain [10], [11].

To address the problem mentioned above, this letter presents a novel FVC estimation method using multispectral images based on deep transfer learning. First, a large amount of training data was simulated by a physical model (PROSPECT + SAIL radiative transfer model, PROSAIL). Second, a pretrained long short-term memory (LSTM) network was built with the simulated training dataset. Then, the network weights of the pretrained model were updated with a limited number of satellite image samples according to the ground measurement. Experiment results were presented for two regions of the Sentinel-2 satellite multispectral images, followed by discussion and conclusion.

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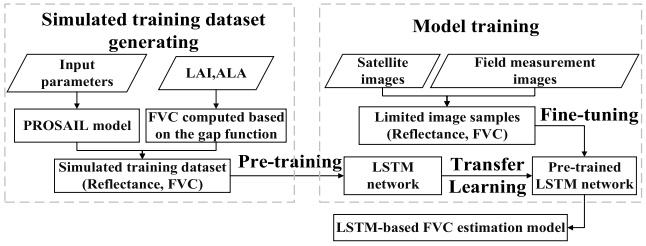


Fig. 1. Flowchart of the proposed FVC estimation method.

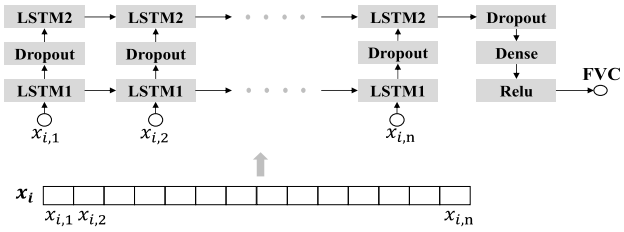


Fig. 2. Diagram of structure of the LSTM-based FVC estimation model.

II. Methods

The flowchart of the proposed FVC estimation method is shown in Fig. 1. The process consisted of two stages: simulated training dataset generating and model training. In the first stage, sufficient simulation data were generated by PROSAIL in the source domain. In the second stage, the LSTM-based estimation model was pretrained with the simulated training dataset. It should be noted that the LSTM was used to extract the vegetation spectral characteristics of the simulated data to establish the relationship model between the simulated reflectance and the FVC. Then, the network weights of the pretrained model were fine-tuned with limited samples from pre-processed satellite reflectance data and FVC extracted from field measurement images in the target domain.

A. LSTM Network

LSTM has recently achieved promising performance with outstanding adaptability for handling sequence data. Its particular neural network structure makes it a DL network with nonlinear fitting ability and robustness stronger than traditional machine learning methods [12]. LSTM is not only used for time series problems, but also in the problem of pixel-level data analysis of remote sensing images. Multiple bands of single or multiple pixels of a single satellite image are input into LSTM at different time steps to extract the spectral or spatial–spectral features of the image [13]–[15]. Therefore, our idea also takes the simulated or satellite multiple spectral bands of a single pixel as the input of the LSTM-based estimation model at different time steps to extract the spectral features. The calculation process of a basic LSTM unit is as follows [16]:

$$C_t = \tilde{C}_t \odot i_t + C_{t-1} \odot f_t \tag{1}$$

$$h_t = g(C_t) \odot o_t \tag{2}$$

where C_t and C_{t-1} are the cell states at different times, $\tilde{C}_t$ is the updated value of the cell state at the current time, f_t is the forget gate, i_t is the input gate, o_t is the output gate, h_t is the output of the LSTM cell in the current time step, the hyperbolic tangent ($g(x) = \tanh(x)$) is used as the activation function, and $\odot$ is the point-wise multiplication between two vectors. For the simulated multiband reflectance input into the FVC estimation model, the cell state C in LSTM structure can record the spectral information of each band, while the unique "gate' structure can be regarded as a screening of the recorded spectral information at each time step and the current information. It means the cell state C of LSTM can effectively extract the features of each spectrum, and its unique "gate" structure can also filter interference information.

Fig. 2 shows the FVC estimation model based on LSTM. A two layers LSTM structure was selected to build the model. And the out of the last time step was the final output of the model after passing through the full connected layer (Dense) and the activating layer (Relu). In addition, a dropout layer was added after each LSTM. And in Fig. 2, $\boldsymbol{x_i}$ was a simulated reflectance vector, $x_{i,1}, x_{i,2}, \ldots, x_{i,n}$ were the values of different bands of $\boldsymbol{x_i}$.

TABLE I
Input Parameters of the PROSAIL Model

Model	Parameters	Units	Value range	Step
PROSPECT	N	-	1-2	0.5
	C_{ab}	$\mu g/cm^2$	20-80	20
	C_{brown}	-	0-0.5	0.5
	C_m	g/cm^2	0.005-0.015	0.005
	C_{ar}	$\mu g/cm^2$	0	-
	C_w	cm	0.005-0.015	0.005
SAIL	LAI	m^2/m^2	0-7	0.5
	ALA	°	30-70	10
	SZA	°	25-55	10
	Hot	-	0.1	-
	VZA	°	2,8	-
	RAZ	°	100,180,280	-

B. Simulated Training Dataset From PROSAIL Radiative Transfer Model

The PROSAIL model is a combination of the PROSPECT leaf optical properties model and the SAIL canopy reflectance model, which was conducted to simulate the reflectance of vegetation canopies. The main input variables of the PROSAIL model are listed in Table I. These input parameters were set in a reasonable range to cover different land cover conditions, so that the training dataset simulated by the model could represent different vegetation types [4], [17]. With reference to the Sentinel-2 data from 2017 to 2019 in the study areas, VZA and RAZ were set to fixed values.

Besides, reflectance values of soils are also the input of the PROSAIL model. In the second step, to remove data redundancy and to reduce computational complexity, 21 representative reflectance spectra were selected from the 47 soil spectra in China obtained from the global soil reflectance database (access at: http://www.isric.org) using the spectral

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angle similarity measurement [18] as following:

$$\alpha_{XY} = \cos^{-1}\left[\sum_{i=1}^{n} x_i y_i \Big/ \left[\left(\sum_{i=1}^{n} x_i^2\right)^{1/2}\left(\sum_{i=1}^{n} y_i^2\right)^{1/2}\right]\right] \quad (3)$$

where X and Y are two different soil reflectance vectors and α_{XY} is the spectral angle of X and Y. The larger the spectral angle, the greater the difference between the spectral vectors. In this study, the spectral angle of 0.05 was used as the threshold to select 21 representative spectra.

Since there is a transforming relationship between leaf area index (LAI) and FVC, the FVC values of the simulated training samples generated from PROSAIL could be calculated by the gap function of LAI and average leaf inclination angle (ALA) [19] using the following:

$$P(\theta) = e^{-\lambda_0 \frac{G(\theta,\theta_1)}{\cos\theta} * \mathrm{LAI}} \quad (4)$$

$$\mathrm{FVC} = 1 - P\left(0^{\circ}\right) \quad (5)$$

where $P(\theta)$ is the gap function, θ is the zenith angle of the incident light ($\theta = 0^{\circ}$ for nadir view), θ_1 is the average leaf angle inclination, $G(\theta, \theta_1)$ is the average projection per unit leaf area, and λ_0 represents the degree of dependence of adjacent layers of leaves ($\lambda_0 = 1$).

To establish the relationship between the simulated database and satellite images, the generated top of the canopy reflectance was computed for each wavelength and then resampled to the multispectral sensor observations according to spectral response profiles [20]. The calculation equation is shown as follows:

$$R = \frac{\int R_{\mathrm{ASD}}(\lambda) \cdot f(\lambda) \cdot d_\lambda}{\int f(\lambda) \cdot d_\lambda} \quad (6)$$

where R is the simulated reflectance values of target bands of the multispectral image, $R_{\mathrm{ASD}}(\lambda)$ is the reflectances simulated by the PROSAIL model, and $f(\lambda)$ is the spectral response profiles of the multispectral sensor. After that, 4 082 400 cases of matched reflectances and FVC values covering various land surface conditions were obtained.

C. Model Training

1) Pre-Training of the LSTM Network: Pre-training is able to improve the generalization performance of DL models because it introduces the prior information of data in source domain into target tasks. Based on this idea, the simulated dataset was used to initialize the parameters of the LSTM network in this letter.

As shown in Fig. 2, in the pre-training process of model training, the multiple spectral band reflectances $(b1, b2, b3, \ldots, bn)$ of a single pixel $\boldsymbol{x_i}$ simulated in the source domain were sequentially input into the LSTM-based estimation model at different time steps. Then the cell state C could record the relevant information of each band and extract the spectral characteristics of different bands. Next, the network parameters were adjusted according to the difference between the estimated FVC and the desired output with iteration of the model. Herein, mean squared error (MSE) was selected as the loss function. Finally, a pretrained LSTM-based FVC estimation model was obtained.

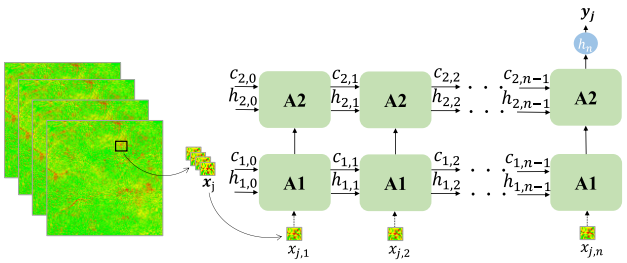


Fig. 3. Diagram of fine-tuning of the pre-trained model.

2) Fine-Tuning of the LSTM Network: After finishing the pre-training, fewer pixels of satellite image reflectance and corresponding FVCs from the field measurement were used to fine-tune the pretrained network. The values of FVC were extracted from field-measured images using the Gaussian fitting and segmentation algorithm [21].

At the fine-tuning process, the network structure of the model was frozen, and the network parameters were fine-tuned. As shown in Fig. 3, A1 and A2 were the two pretrained LSTMs. The $\boldsymbol{x_j}$ was the reflectance vector of a single pixel in the satellite image. And the $x_{j,1}, x_{j,2}, \ldots, x_{j,n}$, which were reflectance values of different bands of $\boldsymbol{x_j}$, were sequentially input into the pretrained model at different time steps. Then, the cell states ($C_{1,0}, C_{1,1}, \ldots, C_{1,n-1}$ and $C_{2,0}, C_{2,1}, \ldots, C_{2,n-1}$) of A1 and A2 could extract the special spectral features of actual satellite data in the target domain, and the output (h_n) of the last time step of A2 was converted to estimated FVC. Next, the difference between the estimated FVCs and the FVC values extracted from the field images were used to update the parameters of the two LSTMs based on back-propagation through time algorithm. Finally, the fine-tuned FVC estimation model was completed.

III. Experiments and Results

We applied the Normalized Difference Vegetation Index (NDVI)-based dimidiate pixel model (DP_NDVI) [5], Levenberg Marquardt-based BPNNs (BP_LM) [4], RF [7], and LSTM, which were only trained by simulated dataset without fine-tuning, and the proposed LSTM-based transfer learning (LSTM_TL) method to the Sentinel-2 satellite multispectral images with 10-m resolution. In the experiments, the satellite images provided by the ESA have been radiometric corrected, geometric corrected, and radiance calibrated. Then the atmospheric correction was conducted using the Sen2cor tool of the ESA to convert the "radiance-at-detector" values into "reflectance-at-surface" data.

Ground measurement data for the experiments were obtained in different areas. Table II shows detailed information of the experiments. Ground sample sites with a size of 20 m × 20 m were collected and FVC of each single sample site was the average value of the FVC extracted from five digital pictures on the diagonal and the center of the sample site. The geographic coordinates of the center of these sample sites were recorded by a handheld global positioning system receiver (Garmin eTrex 209X). Besides, the sample sites of the experiments were divided into 7:1:2 for training, validating, and testing.

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TABLE II
GROUND SAMPLING OF THE EXPERIMENTS

No. of the Study area	Location	Vegetation types	Sampling date of ground samples	Number of ground samples	Acquisition date of Satellite images
I	Xiong'an New Area	wheat	Nov.7-9, 2020	121	Nov.8, 2020
II	Beijing	wheat, pine, grass	Mar.25,2021; Apr.1-2, 2021	134	Mar.23,2021; Apr.4, 2021

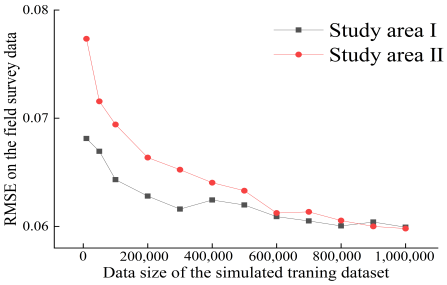


Fig. 4. RMSE of the proposed LSTM-TL method on the field survey data under different data size of the simulated training dataset.

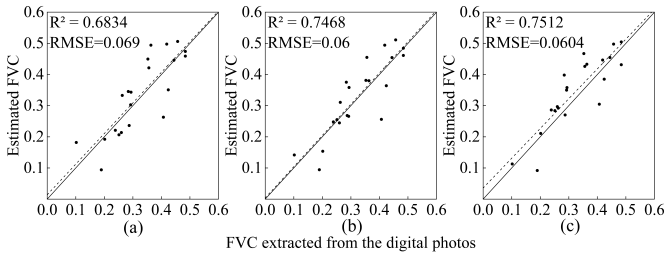


Fig. 5. Scatter plots of FVCs estimated by the proposed LSTM-TL method and that extracted from the digital photos using (a) two bands, (b) three bands, and (c) four bands in experiment I.

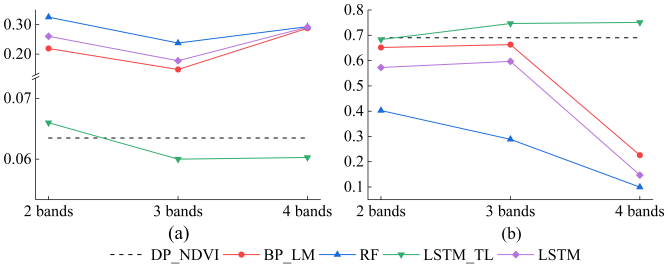


Fig. 6. Comparisons of the (a) RMSE and (b) R^2 of different FVC estimation models with different bands combination for single vegetation type in experiment I.

To evaluate the performances of the above methods, the root-mean-square error (RMSE) and the coefficient of determination (R^2) were calculated from the estimated FVC and true FVC extracted from field measurement [17]. In order to improve the training efficiency and accuracy, the network parameters of the proposed FVC estimation model were determined by trial and error. The number of hidden neurons in the LSTM unit was set to 100, and the drop rate of each dropout layer was set to 0.3. The objectives of these experiments were: 1) to confirm the data size of simulated training dataset required to pre-train the source model; 2) to find the optimal subset of bands among different band combinations, two bands (red and near-infrared), three bands (red, near-infrared, and green), and four bands (red, near-infrared, green, and blue); and 3) to make a comparison on the effectiveness of single and multiple vegetation types in different regions when comparing with the DP_NDVI, BP_LM, RF, and LSTM algorithms.

Fig. 4 shows the RMSE of the proposed LSTM_TL method on the field survey data under different data size of simulated training dataset when using four bands data. The RMSE of the proposed method decreased with the increase of the data size, and tended to be stable when the data size reached 8 00 000. Therefore, to keep consistent with the experimental conditions of the machine learning methods in references [4], [7], 9 00 000 cases of matched simulated reflectances and FVCs were randomly selected, and it was assumed that the source model could be pretrained properly using a large training data size.

Comparison using field survey data was carried out to evaluate the performance of the proposed FVC estimation algorithm for the Sentinel-2 multispectral data. The scatter plots of the estimated FVCs using different number of bands and those calculated from the field photos in the ground survey sites are shown in Fig. 5. It is clearly observed that there is high spatial consistency between the estimated FVCs and measured FVC values. The estimated FVC values of the proposed approach using different bands combination were mainly concentrated in 25–50%. This was a reasonable distribution because the ground sample sites were covered with wheat at the three-leaf stage and the pre-tiller stage, which was relatively short and sparse. Thus, this direct comparison could confirm the reasonability and reliability of the proposed LSTM_TL method.

Fig. 6 shows the (a) RMSE and (b) R^2 of different FVC estimation models with different bands combination in experiment I, which was only covered with single vegetation type (wheat). All the approaches yielded relatively good estimations using three spectral bands. It should be noted that when the proposed method was used to estimate FVC, the accuracy of three bands (RMSE = 0.06, R^2 = 0.7468) was comparable to four bands (RMSE = 0.0604, R^2 = 0.7512). Besides, the performance of the proposed LSTM_TL was always better than that of LSTM, indicating that a small size of image samples can significantly improve the estimation accuracy of LSTM by learning spectral features of actual satellite images to decrease the uncertainties of the pretrained model.

Various vegetation types, including wheat, pine, and grass, were considered in experiment II. As shown in Fig. 7, the

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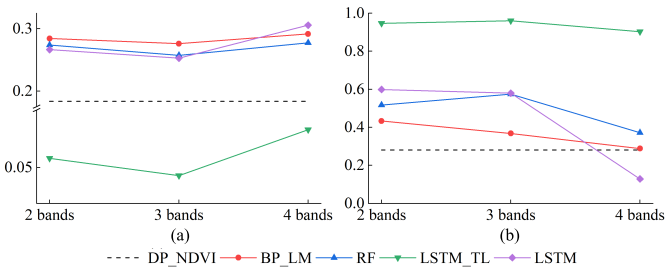


Fig. 7. Comparisons of the (a) RMSE and (b) R^2 of different FVC estimation models with different bands combination for three vegetation types in experiment II.

overall accuracy of DP_NDVI showed a significant decline (RMSE increased by 181% and R^2 decreased by 59%) when compared with the single vegetation type in experiment I. The performance degradation of LSTM, BP_LM, and RF methods was comparatively small. Meanwhile, the proposed method achieved the highest overall accuracy using three bands (RMSE = 0.0443 and R^2 = 0.9605). This is because green spectral band contributes to the biophysical variability of vegetation, while blue band is affected by the residual after atmospheric correction.

IV. Discussion and Conclusion

This letter proposed a novel FVC estimation method based on pretrained LSTM with a large number of simulated training samples generated from PROSAIL radiative transfer model. A small amount of ground samples which were extracted from the field digital pictures and corresponding reflectance data of Sentinel-2 satellite multispectral images were used to fine-tune the pretrained estimation model. The experiments were conducted to confirm the data size of simulated training dataset and to evaluate the performance of the proposed LSTM_TL method over two regions and multiple vegetation types.

In this study, part of the simulated training data could guarantee the performance of the proposed LSTM_TL method, which means that the source model could get stable generalization ability under 8 00 000 training samples. The proposed LTSM_TL method significantly outperformed the DP_NDVI, BP_LM, RF, and LSTM, achieving much better accuracies and estimations in the two experiments. When modeling with three spectral bands, the performance of the proposed LTSM_TL was comparable to or even better than that of four bands. Therefore, it is a valuable choice to establish FVC estimation model using three bands of Sentinel-2 data. Meanwhile, the two experiments in different study areas estimated the coverage of single and multiple vegetation types. The proposed technique could overcome the difficulties in determining the parameters in the empirical method and machine learning methods for FVC estimation, which are usually varied with vegetation types and regions. In addition, the most important advancement of this study is that the proposed method combines LSTM and transfer learning which can greatly improve the FVC estimation accuracy of LSTM with limited image samples data. All the above analyses confirm the remarkable estimation performance and strong robustness of the proposed method.

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